

Image Recommendation Based on Affective Neural Networks with Graphs

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Abstract

Nowadays, there is an exponential increase in internet consumption, from searching for information and visiting websites to using social media or creating content. All of this also involves constantly viewing images that are displayed while scrolling. We can easily associate emotions with images, in the sense that any image triggers unique and exclusive emotions in the user. Most of these images are recommended to users through Recommender Systems that believe they are the best choice for the user in question, in order to help them find information that is relevant to them and avoid unwanted image consumption. Therefore, together with the work that has already been developed, from the creation and construction of the database to the creation of three collaborative filtering recommendation systems, we intend to develop a hybrid recommendation system based on emotions, using graph methods, in order to make recommendations more effective and accessible.

Keywords: Recommended Systems; Graphs; Images; Emotions; Hybrid Systems

1 Introduction - POR MELHORAR

In one way or another, everything in life is based on visualizations. Nowadays, with the growth of internet consumption and a generation much more inclined to consume online content, viewing countless images while scrolling is extremely common and directly interferes with the user experience.

When viewing an image, a wide range of emotions are triggered in humans. The emotion that an image can bring is extremely volatile for each user, being something unique and personal—the triggering of an emotion depends on several factors, such as how the user views it, the environment they are in, their past experiences and associated memories, or even their hobbies. Taking all this into account, it is understood that the images suggested to each user must undergo an intense personalization process, which is done by recommendation systems, and this is what this study explores.

The more refined, personalized, and adapted to images and emotions this system is, through critical analysis and the creation of possible correlations between images and feelings triggered, as well as user characteristics, the better the user experience will be. Thus, the main objective of this

thesis focuses on the use of a hybrid distribution system, as it allows us to deal with the high computational cost of visual processing, the complexity inherent in the emotional side, the need for low latency in affective personalization, and the scalability necessary to handle multiple models and large numbers of users and content.

To evaluate the effectiveness of the hybrid system I propose, its results will be compared with three alternative approaches: a system based exclusively on emotions integrated into a collaborative filtering model, a system that uses only demographic information about users, and finally, a system of recommendations generated entirely at random, work that has already been done within the same project.

2 Background

2.1 Human Emotions: Definition, Characterization, and Theoretical Models

There is no universally accepted definition, but we have several characterizations. Paul Ekman defines emotions as “automatic psychophysiological responses to events that have meaning for an organism” [1], while Richard Lazarus argues that emotions result from the cognitive evaluation of an event—that is, the interpretation of its significance for well-being [2]. Scherer proposes a component-based definition, according to which an emotion results from the coordination of multiple subsystems: cognitive evaluation, preparation for action, motor expression, physiological response, and subjective experience [3].

Thus, it is safe to consider that emotions are rapid psychophysiological reactions triggered by conscious or unconscious evaluation of a stimulus, and that they influence behavior, physiology, facial expression, and subjective experience.

Understanding how emotions are triggered in humans is equally essential, and research suggests that they generally arise from three complementary mechanisms. First, we have appraisal models, such as the one proposed by Lazarus, which argue that the brain continuously evaluates 1) the relevance, 2) novelty, and 3) meaning of stimuli, followed by an emotional response, taking into account whether the evaluation attributes significance to the stimulus or not [2]. At the same time, Ekman and LeDoux defend a perspective of automatic and biological mechanisms, proposing that many

emotions are activated by rapid and automatic circuits, especially in the amygdala, before conscious processing occurs [1, 4]. Finally, contextual and cultural perspectives, such as those proposed by Russell, argue that the emotional meaning of a stimulus depends on cultural norms, social expectations, and lived experiences [5].

Regarding the characteristics of emotions, science converges on a set of main characteristics that allow them to be easily distinguished from other affective states. These characteristics highlight emotions as brief and intense phenomena, guided by physiological, expressive, cognitive, behavioral, and subjective components [3]. This allows emotions to play an adaptive role, facilitating processes such as decision-making, social communication, and behavioral regulation [6].

With regard to the typology of emotions, two models stand out. The first is proposed by Ekman and defines emotions as discrete entities, identifying a set of basic emotions—joy, sadness, anger, fear, surprise, and disgust [1]. The second model was proposed by Russell and defines emotions using two fundamental axes—valence (pleasant—unpleasant) and activation or arousal (calm—excited) [5]. This latter model has been very relevant in the technological field, as it is easy to represent mathematically and is compatible with machine learning techniques. Understanding these emotional models is crucial in the context of emotion-aware recommender systems, as the affective impact of stimuli—such as images—significantly shapes user preferences, choices, and interactions. Integrating emotional responses, whether explicit or implicit, enables the development of more human-centered systems that are sensitive to affective states and potentially more effective in personalizing content.

2.2 Recommender Systems

2.2.1 Definition and Functioning.

Recommendation systems play an important role in filtering information, something that is increasingly necessary today. They help users discover items of interest from large volumes of information, with the aim of reducing information overload, acting as mediators between users and large volumes of content. The classic definition describes these systems as tools that “provide personalized suggestions” based on users’ explicit or implicit preferences [7]. It is also worth noting that these systems have the central objective of predicting the usefulness of an item for a given user, using statistical models or machine learning methods [8]. The way they work is based on user habits, item characteristics, user attributes, and the collection and analysis of different types of data, whether explicit information—provided directly by the user through assigned ratings—or implicit information—clicks and viewing time/history. By analyzing this data, the system can capture preference patterns and estimate the user’s future interest in new items. There is a conceptual framework [8] that defines this process in three fundamental

steps (1) user representation; (2) item representation; and (3) matching algorithm between the two.

2.2.2 Main Paradigms.

There are three main paradigms of recommendation systems. The first is Collaborative Filtering, a system based on the idea that users with similar behaviors or preferences tend to like similar items. According to GroupLens—one of the leading large-scale collaborative filtering systems—the method does not depend on textual content, but rather on user ratings [9]. In this way, the system is able to reflect personal preferences and identify patterns of similarity between users, recommending items that are highly likely to be relevant to each individual. Instead of promoting items just because they are popular, collaborative filtering adapts to the specific tastes of each user, offering a more effective mechanism for navigating and prioritizing information in large-scale environments. This filtering can be divided into memory-based and model-based methods, the most common in large-scale environments [10]. The second paradigm is Content-Based Filtering, where systems define user preferences through the items themselves, thus building user profiles [8] based on attributes such as text or metadata. The main advantage of these methods is their independence from other users’ data. The third paradigm represents hybrid systems, which ultimately combine multiple sources of information. These systems can be classified into seven categories [11]—weighted, switching, mixed, feature combination, feature augmentation, cascade, and meta-level—and arise in response to problems such as cold start, sparsity, and lack of diversity [10].

2.2.3 Other Approaches and Emotion Approaches.

In addition to these three main paradigms, with the advancement of deep learning techniques, approaches based on Deep Learning [12] have emerged and, in parallel, the incorporation of contextual factors has led to the development of context-aware recommender systems [13].

In the field of emotion-based recommendation, the field of Affective Computing, introduced by Picard [14], stands out. These systems use emotional information as additional context to improve personalization, fitting naturally into the evolution predicted by Adomavicius and Tuzhilin [8] towards multidimensional recommendation systems capable of integrating multiple modalities and contextual signals.

2.3 Graphs

Graphs are a fundamental mathematical structure for representing entities and the relationships between them. A graph consists of a set of nodes (or vertices) and a set of edges that define how these nodes are connected. They are suitable for modeling complex systems in which the interactions, dependencies, or similarities between elements are as important as or more important than the elements themselves. Due to their versatility, graphs have been widely used in

areas such as social network analysis, information retrieval, computational biology, and recommendation systems [15].

Graphs can be classified into different types according to the nature of their connections and properties [17, 18, 19]:

- **Directed Graphs:** the edges have a direction, indicating asymmetric relationships between nodes. For example, in a social network, a connection “follows” from one user to another.
- **Undirected Graphs:** edges have no direction, representing symmetric relationships, such as friendship or similarity between images.
- **Weighted graphs:** each edge has a weight, which can represent the intensity of the relationship, such as the strength of a user’s preference for an image or the emotional similarity between two images.
- **Heterogeneous Graphs:** contain different types of nodes and edges, allowing the modeling of complex relationships between diverse entities, such as users, images, emotions, and visual characteristics.
- **Multimodal Graphs:** an extension of heterogeneous graphs, where multiple data modalities (visual, textual, emotional, contextual) are integrated into a single structure.

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An important point to note is that the modeling of user-image-emotion relationships is best represented through heterogeneous or multimodal graphs, which allow visual, affective, and demographic information to be combined into a single graph, serving as the basis for recommendation algorithms and Graph Neural Networks.

Graphs have become central to recommendation systems due to their ability to naturally represent heterogeneous and multimodal relationships. In addition to their representational power, graph-based methods enable the propagation of information across the network, including the transmission of user preferences between similar items, the transfer of emotional characteristics between visually related images, and the integration of demographic or contextual information. These capabilities are fundamental in hybrid recommendation systems, which benefit from combining multiple data sources into a unified structure.

The recent development of Graph Neural Networks (GNNs) has significantly strengthened the role of graphs in machine learning. GNNs extend deep learning techniques to graph-structured data by iteratively aggregating information from neighboring nodes [16].

In the field of recommendation systems, graph-based methods have achieved state-of-the-art performance by leveraging these advanced neural architectures. Knowledge graph-based approaches, such as the KGAT model proposed by Wang et al. [17], have shown how semantic relationships between entities can improve the accuracy and interpretability of recommendations. More broadly, Shi et al. [18] present an

extensive review on the use of GNNs in recommendation systems, describing how graph structures allow the integration of user–item interactions, content features, and contextual information into a unified neural model.

Despite their advantages, graph-based systems present challenges. Large-scale graphs require significant computational resources, especially when training deep GNN models. Other problems include data sparsity, graph noise, and risks such as over-smoothing—a situation in which node representations become indistinguishable after multiple layers of propagation. In addition, distributed storage and computing are often required for graphs representing millions of users and items. Nevertheless, the ability of graphs to unify heterogeneous information, model relational structures, and support deep neural processing makes them particularly well-suited for applications involving affective personalization and multimodal content.

2.3.1 Neo4j: Graph Database Management System.

Neo4j is a graph-oriented database management system widely used in applications that require the explicit representation of entities and their relationships. Based on the property graph model, Neo4j stores information in nodes, edges, and associated properties, allowing the modeling of complex structures in an intuitive and expressive manner. One of its main features is the Cyphe query language, developed to facilitate graph navigation thru relational patterns, which makes the execution of operations such as neighbor search, path calculation, and connectivity analysis efficient.

In addition to its expressiveness, Neo4j stands out for its performance in highly relational queries, where traditional table-based systems tend to exhibit limitations. Its graph-optimized architecture allows for low latency in traversal operations and offers indexing and caching mechanisms adapted to non-linear query patterns. Neo4j also includes support for analytical processing thru the Graph Data Science module, which provides centrality, community, similarity, and embedding creation algorithms, allowing for advanced structural analyzes directly on the graph.

In this way, Neo4j constitutes a robust tool for the construction and analysis of graph databases, supporting use cases such as social networks, recommendation engines, fraud detection, knowledge analysis, and modeling of complex systems.

2.4 Emotions related to Images

Images are visual stimuli that can provoke immediate emotional responses in individuals. The resulting emotional experience is highly subjective, varying according to individual factors such as memories, culture, and context, as well as intrinsic characteristics of the image itself, such as color, shape, and composition. Understanding these relationships is fundamental in the context of this thesis, as the affective personalization of image recommendations depends on the

ability to model how different visual stimuli trigger emotional responses in users.

The processing of emotions induced by images involves multiple cognitive and neural levels. During initial visual perception, the brain quickly processes basic patterns, including shapes, colors, and contrasts. Areas such as the amygdala respond to emotionally significant visual stimuli, often before conscious processing occurs [1]. Parallely, cognitive evaluations, conscious or subconscious, interpret the meaning of the image and its context, modulating the emotional response [2].

Various visual characteristics are determinants of the intensity and type of emotion evoked. Color plays an important role: warm tones tend to generate excitement or energy, while cool tones often induce calmness or sadness. Brightness and contrast can amplify or attenuate emotional response, while composition and visual symmetry influence esthetic and emotional perception; balanced images tend to provoke positive responses, while chaotic compositions can generate tension or discomfort. Semantic content is also relevant: faces, familiar objects, or threatening scenes activate specific emotional responses, often universally [3].

To model the relationship between images and emotions, there are different theoretical and computational approaches. Discrete models categorize emotions into basic classes, such as joy, sadness, fear, anger, surprise, and disgust, following Ekman's principles [3]. On the other hand, continuous models use dimensions such as valence (pleasant-unpleasant) and activation/arousal (calm-excited), as proposed by Russell [4]. In the computational domain, visual feature extraction techniques, deep learning embeddings, and Graph Neural Networks (GNNs) allow for predicting the likelihood of an image inducing a specific emotional response. Incorporating this information into recommendation systems allows for the construction of emotional profiles of users, where each interaction with an image contributes to a personalized affective map. When integrated into distributed hybrid systems, these representations enable recommendations that combine content similarity, interaction history, and expected emotional impact. This type of affective personalization significantly increases user relevance and satisfaction, as the recommendations are not limited to objective criteria but are also sensitive to the emotional impact of the images.

Despite the advantages, there are challenges associated with this approach. Subjective variability implies that the same image can generate distinct emotional responses in different users, which complicates the generalization of predictive models. In addition, the extraction and calculation of emotional features for large volumes of images is computationally expensive, requiring infrastructures capable of handling this load. Finally, it is necessary to ensure that the data used to train the emotional prediction models are high quality, accurate, and representative of the target audience.

refs. Não podes só colocar informação sem referências

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Isto como é difícil estudos que tenham tudo, pode ser dividido em:

* Sistemas de recomendação de imagens / multimédia

3 Related Work

3.1 Emotional Analysis of Images

The emotional analysis of images plays a central role in understanding human perception and in the computational modeling of affective responses. Research in this area is summarized by databases of visually annotated stimuli, among which the International Affective Picture System (IAPS) [20] stands out, being one of the most widely used resources in research on emotions, psychophysiology, and neuroscience. The IAPS provides a diverse set of images that include various categories – nature, violence, eroticism, natural disasters, everyday activities, animals, neutral objects – evaluated according to Russell's two-dimensional model (valence-activation) [21], allowing for the mapping of the average affective response of the population to visual stimuli. This instrument has become a reference as it ensures standardization between studies, avoiding cultural or contextual variation of stimuli, allows scientific reproducibility, and serves as a basis for benchmarking computational models of emotional prediction. There are studies that prove that the IAPS triggers consistent psychophysiological responses – such as heart rate measurement and skin conductance – which validates its effectiveness as a scientific tool [20]. More recently, the Open Affective Standardized Image Set (OASIS) [22] expanded this approach, offering large-scale collected annotations, with modern evaluation methodologies and greater content diversity.

With the advancement of deep learning, the emotional analysis of images has evolved significantly. Before, methods based on manual features were used, but today most solutions use fully learned models. Convolutional networks like VGG and ResNet have proven to be very effective in extracting important visual patterns, serving as a foundation for models that attempt to predict emotions. One of the most well-known works is EmoNet, developed at MIT [23]. This model receives an image, extracts its deep visual representations, and directly transforms them into emotional categories or valence and activation values. The study showed that networks trained for visual tasks can naturally learn patterns that correspond to the emotions people feel when viewing an image. At the same time, several studies have explored the prediction of valence and arousal using both CNNs – Convolutional Neural Networks, a type of deep neural network especially used for image processing – and Vision Transformers, which have achieved even better results [24]. In many cases, these models are trained with transfer learning, which allows for good performance even when there are few annotated data. Overall, research shows that it is possible to predict the emotion of an image with good accuracy based on its high-level visual features, which reinforces the link between visual composition and human emotional response.

Despite these advances, most studies focus on predicting emotions in isolation, not exploring the integration of these

isto pode ir para uma secção de data, não é related work

responses into recommendation systems or complex data structures like graphs. Moreover, many models assume generalizable emotional responses, when in reality emotional perception is strongly influenced by context, personal experience, and cultural differences. These limitations highlight the need for systems capable of combining automatic emotional analysis with contextual personalization — a gap that directly motivates the objectives of this project.

3.2 Emotion-Aware Recommender Systems ✓

Traditionally, Recommendation Systems are based on behavioral patterns, explicit preferences, or content characteristics. However, the literature has shown that users' emotional response plays an essential role in how content is perceived, consumed, and evaluated. With the natural evolution of the field of recommendation processes, along with the continuous development of the area of Affective Computing, introduced by Picard [14], the integration of affective signals into recommendation systems emerged, highlighting the importance of computational systems capable of recognizing, interpreting, and responding to human emotions. ^{ref}

Regarding recommendations, the integration of emotion can occur in two main ways: (1) user emotional modeling and (2) content emotional modeling. In the first, the systems seek to determine the user's emotional state - thru self-report, ^{through} facial expressions, physiological signals, or interaction patterns - so that a well-adapted recommendation to the state in question can be possible. ^{le preciso referencias para isto tudo} The second aspect focuses on the emotion conveyed by the content itself, being a type of system that uses computational models capable of identifying emotions present in images (3.2.1), videos, or music, often utilizing deep representations extracted by CNNs or Vision Transformers, as discussed in the previous section. ^{3.2.1} ^{???} ^{refs}

Additionally, there are hybrid approaches that combine the two methods explained earlier. These approaches have been showing improvements in personalization, but they remain limited by factors such as the scarcity of emotion-annotated datasets, high individual variability in affective responses, and difficulty in capturing the emotional context in real-time. Moreover, many of these systems treat emotions as independent attributes and do not explore complex relational structures between users, content, and emotions. ^{refs}

Thus, despite the advances that have occurred, the vast majority of emotional recommendation systems remain disconnected from a good data structure, such as graphs, and rarely benefit from deep relational information or scalable models used in real-world scenarios. This limitation opens up space for new systems that combine automatic emotional analysis, contextual personalization, and structured representations — a gap that the present project seeks to fill through the use of graph models and a hybrid emotion-based recommendation approach. ^{temos de explicar como é que vamos fazer o híbrido}

3.2.1 Emotional Analysis in Images.

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The emotional analysis of images is one of the most established domains within affective processing, benefiting from the high maturity of computer vision models.

The process usually includes three stages: (1) extraction of deep representations from the image, (2) projection of these representations into an affective space, and (3) aggregation of the result into metrics usable by a recommendation system. Databases such as IAPS [20], OASIS [22], and EMOTIC [24] provide visual stimuli with standardized emotional annotations, allowing for the training of robust and comparable models across studies.

When integrated into recommendation systems, these models allow for inferring the likely "emotional impact" of images associated with products, movies, or digital content. This makes it possible to adjust recommendations according to the user's current emotional state, avoid unwanted stimuli, or select content visually compatible with their affective preferences.

3.3 Visual Content Recommender Systems ^{saiu}

Visual content-based recommendation systems came to counter the limitations of traditional methods that relied solely on metadata or behavioral patterns. With the development of computer vision techniques and deep learning models, it has become possible to extract rich visual representations, capable of capturing texture, composition, color, style, and high-level semantics present in images and videos. These representations enable the creation of visual profiles for users and items, something especially useful in domains such as fashion, art, photography, social media, and e-commerce. ^{REFS} Initially, these systems used manual descriptors, such as SIFT[25] or manual color histograms, which had certain limitations in capturing semantic meaning. The introduction of CNNs, such as VGG[26] and ResNet[27], revolutionized this area by providing deep visual embeddings capable of delivering high-quality semantic information. These advances led to the creation of recommendation methods based on visual similarity and supervised models that learned relationships between user preferences and visual characteristics of items [12].

~~There are notable works in this domain, specifically the visual recommendation systems for fashion introduced by McAuley [28], which have proven that embeddings extracted from CNNs can effectively model visual styles. Subsequently, the VBPR model – Visual Bayesian Personalized Ranking [29] integrated visual embeddings directly into the personalized ranking process, which demonstrated significant gains in cold start scenarios and in domains where appearance is critical. More recently, Vision Transformers (ViTs) have expanded this capability by allowing the representation of global spatial relationships thru attention mechanisms, often surpassing CNNs in complex visual tasks. Recent studies in the multimodal and visually oriented domain of recommendation, described in surveys such as that of Zhang et~~
^{trabalho de 2015, muito antigo para RW. De certeza que ha trabalhos melhores}
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al. [12], show that end-to-end models combining images, interactions, and context achieve superior performance in preference prediction, personalized ranking, and item retrieval. Some practical applications of these systems can be the recommendation of visually similar products (fashion, furniture, decoration), discovery of artistic styles, selection of thumbnails, and recommendation on inspirational platforms. The main advantage of visual content-based systems is their ability to effectively handle the cold start problem — allowing the recommendation of new items based solely on their appearance.

However, challenges remain. Visual representations do not always capture contextual, cultural, or emotional nuances. Moreover, many visual-based models ignore the relational structure between users, items, and visual attributes, information that can be explicitly represented thru graphs or hybrid models. These limitations reinforce the need to integrate visual information with relational structures and affective models. tenho a certeza que existem mais estudos e mais info sobre sistemas de recomendação "visuais", de imagem conteúdo multimedia

3.4 Graph-Based Recommender Systems

Graph-based recommendation systems have enabled the explicit modeling of relationships between users, items, and auxiliary information in a structured and flexible manner. In these systems, elements such as users, content, categories, or emotions are represented as nodes, while interactions, similarities, or semantic relationships are encoded as edges. This representation faithfully reflects the relational nature of recommendation problems. Regarding network science, a recommendation system can be interpreted as a heterogeneous graph, composed of various types of nodes and their respective relationships, which allows for the direct application of graph theory concepts and complex network analysis [15].

3.4.1 Graph Neural Networks for Recommendation Systems.

já que as GNN são importantes, fala mais sobre isso no background, podes colocar até esquema da arquitetura
Graph Neural Networks (GNNs) só definir a primeira vez que se usa have become one of the **most relevant paradigms in** the field of graph-based recommendation systems. These networks promote vector representations of nodes thru an iterative process of aggregating information from their neighborhoods, simultaneously combining the graph structure and the attributes of the nodes [16]. In the context of recommendation, GNNs facilitate the capture of complex and high-order relationships between users and items, something that overcomes many limitations of traditional methods. Superior to that, knowledge graph-based approaches, such as the Knowledge Graph Attention Network (KGAT), integrate explicit semantic relations and attention mechanisms that promote the relative importance of different types of connections [17]. Recent review studies indicate that GNN-based recommendation systems show significant performance gains, especially in scenarios involving heterogeneous data, rich in relationships and contextual

information [18]. These characteristics make graphs particularly suitable for modern recommendation applications, where users, content, and contexts are strongly interconnected. falta efetivamente falar de estudos recentes que usem grafos para fazer as recomendações.

3.4.2 Advantages and Limitations.

The advantages lie in a natural and expressive representation of the complex relationships between entities and the possibility of exploring indirect connections and higher-order relationships, enriching the recommendation process. Moreover, graphs facilitate the integration of multiple sources of information—such as visual, contextual, or emotional characteristics—into a single structured model [15][18]. Another important benefit is the potential increase in system transparency, as the paths in the graph can be analyzed to explain recommendations.

There are also limitations to consider, such as the construction, maintenance, and updating of large-scale graphs, which can be computationally expensive, especially in dynamic environments. Models based on GNNs also face scalability issues and require care to avoid phenomena such as over-smoothing and loss of discriminative information [16]. Additionally, most current systems focus primarily on structural relationships and interaction patterns, neglecting affective factors such as emotions. When these are considered, they tend to be modeled as static attributes, ignoring their dynamic and context-dependent nature—something that highlights the need for hybrid approaches, using graphs to simultaneously represent users, visual content, and emotional responses.

3.5 Hybrid Approaches Combining Emotions, Visual Features and Graphs

Over time, recommendation systems have notably evolved, and it has become evident that isolated approaches, based solely on interaction patterns or visual content, are insufficient to capture the complexity of human behavior. To address this, hybrid approaches have been developed that seek to combine multiple sources of information, such as visual characteristics, emotional states, and relational structures, with the aim of improving the quality, personalization, and robustness of recommendations [11][12]. These hybrid systems were initially proposed as a way to minimize or mitigate classic limitations of collaborative and content-based filtering, such as data sparsity and the cold-start problem[11]. In parallel with the development of deep learning and affective analysis, these approaches have also evolved, enabling the integration of emotional information and deep visual representations, opening doors to systems more sensitive to the context and subjective preferences of users.

Recently, the integration of emotions into recommendation systems has been addressed, mainly thru the incorporation of affective attributes automatically extracted from the content. Regarding visual content, models based on CNNs

or Vision Transformers allow the extraction of high-level embeddings that, in addition to capturing the semantics of the image, also capture cues related to its emotional load [23][24]. Visual and emotional embeddings are used as additional features in recommendation models, in order to align the recommended content with the user's emotional state or their long-term affective preferences. Works in this area demonstrate improvements in the perceived relevance of recommendations, especially in domains such as multimedia, social networks, and entertainment platforms [14]. However, many of these approaches treat emotions as static attributes of the content, without exploring their dependence on context or the inter-individual variability in emotional perception. Furthermore, the combination of visual and emotional characteristics is often carried out in a vectorial manner, ignoring more complex structural relationships between users and items.

In order to complement what has been mentioned above, graphs emerge to offer a natural framework for representation between users, content, emotions, and contexts. Within hybrid systems, based on graphs, all these entities can be modeled as interconnected nodes, allowing visual and emotional information to be reproduced thru the relational structure of the system [15]. Recent approaches based on GNNs demonstrate that learning representations in heterogeneous graphs allows for the capture of higher-order interactions between different types of entities, something that surpasses models that operate solely on independent embeddings [16][18]. Particularly, knowledge graphs applied to recommendation allow for the integration of semantic, emotional, and contextual attributes in a unified manner, as demonstrated by models such as KGAT [17].

Despite these advances, the explicit integration of emotions automatically inferred from visual content into recommendation graphs is still limited. In most cases, emotional information is treated as an additional node attribute, without exploring its temporal dynamics or its relational influence in the graph.

3.6 Discussion

The state-of-the-art analysis demonstrates that there are significant advances in the area of recommendation systems. However, regarding the incorporation of emotions, the use of deep visual features, and the adoption of graph-based models, these developments have largely occurred in a fragmented manner. Most existing systems integrate only one or two of these dimensions, and rarely explore their combined potential in a systematic and coherent manner. More specifically, emotional information automatically extracted from visual content is, in most cases, treated as a static and isolated attribute, without considering its subjective, dynamic, and context-dependent nature. Similarly, many emotional recommendation systems do not explore rich relational structures, limiting themselves to tabular or vector models that do not

adequately capture the complex interactions between users, content, and affective states. Although graphs have demonstrated high effectiveness in modeling these relationships, their integration with deep emotional analysis and contextual personalization remains limited in the literature. Moreover, despite the growth of hybrid approaches, few models can simultaneously integrate automatic emotional analysis, deep visual features, and graph structures into a single scalable framework. The computational complexity associated with multimodal fusion and the scarcity of emotionally annotated datasets continue to be relevant challenges, hindering the application of these systems in real-world scenarios [12][18].

These limitations highlight a clear gap in current research and motivate the need for recommendation systems that integrate emotional analysis of images, contextual personalization, and graph-based relational modeling. The convergence of these dimensions allows for the representation not only of the recommended content but also of how it is emotionally perceived and its relevance to a specific user in a specific context. In this sense, hybrid approaches that use graphs as a central structure to integrate users, visual content, and emotions emerge as a promising direction for the development of more expressive, interpretable, and adaptive recommendation systems — a framework that serves as the foundation for the proposed work.

podes criar uma secção de Data e falar dos datasets que existem, o IAPS e assim, e falar do nosso dataset nessa secção. Podes citar as teses da Inês e da Leonor como tendo sido o inicio da data collection e falar dos trabalhos delas no RW

4 Methodology and Work Plan

4.1 Data Analysis

The first stage of this work consists of a detailed analysis of the dataset used in the project. The dataset, created later and constantly developed by the research team, contains information related to users, images, emotional responses, and other contextual and demographic attributes. At this stage, an exploratory analysis is conducted with the aim of understanding the structure, dimension, and characteristics of the available data. This analysis includes the identification of the types of emotional responses collected, the number of annotations per image and per user, as well as the distribution of interactions throughout the dataset. Particular emphasis is placed on the relational nature of the information, namely the multiple associations between users, images, and emotions, which naturally justify the adoption of a graph-based representation.

>> COLOCAR ANÁLISE DA BD COM TEXTO PRINCIPAL E DEPOIS RELAÇÕES E GRÁFICOS RELEVANTES

A análise dos dados para um capítulo de preliminary results

4.2 Graph Database Development

Once the data analysis is complete, the dataset will be modeled and stored in a graph-oriented database. The objective of this stage is to represent users, images, emotions, and their interactions in the form of nodes and relationships, allowing

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for an efficient modeling of complex and heterogeneous relational structures. The database will be implemented using Neo4j, which offers native support for property graphs and expressive query mechanisms thru the Cyphe language. In the proposed model, users, images, and emotions will be represented as distinct node types, while the relationships will encode interactions such as image viewing, emotional annotations, similarity, or affinity between entities.

4.3 Hybrid Graph-Based Recommender System

The final stage of the methodology focuses on the implementation of a hybrid recommendation system based on the graph representation of the data. The proposed system aims to combine emotional information inferred from visual content with data related to users and the relationships existing in the graph. The recommendation process will integrate multiple signals, including the emotional similarity between images, user preferences, and measures of proximity or connectivity in the graph. Hybrid strategies will be explored to reconcile emotional relevance with personalization and the diversity of the recommendations produced. For evaluation purposes, the dataset will be divided into training and testing subsets, using appropriate separation strategies that preserve the temporal and relational characteristics of the data. The performance of the proposed system will be compared with benchmark approaches. The performance of the proposed system will be compared with benchmark approaches. The evaluation will be based on standard metrics of recommendation systems, including precision-oriented metrics and ranking metrics, as well as a qualitative analysis of the level of personalization and relevance of the recommendations.

4.4 Summary

In summary, the proposed methodology combines data analysis, graph-based modeling, and hybrid recommendation techniques to address the limitations identified in the literature. By explicitly exploring the relational nature of emotions and visual content, this approach aims to contribute to the development of more expressive, scalable, and emotionally context-sensitive recommendation systems.

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5 Forthcoming Work and Conclusions

The next stages of this project will focus on the consolidation and implementation of the proposed methodology. Once the exploratory analysis of the dataset is deepened, a graph database will be developed to structurally represent the users, visual content, and associated emotions, as well as the relationships between these entities. Subsequently, a hybrid recommendation system based on graphs will be implemented, which will be evaluated using metrics suitable for recommendation systems, allowing for the analysis of the impact of integrating emotional information and graph structure on overall performance.

So far, the project has allowed for an in-depth review of the state of the art, identification of relevant gaps in the literature, and the definition of a methodological approach consistent with the proposed objectives. A preliminary analysis of the data was also conducted, providing a solid foundation for the subsequent phases. The main difficulties identified are related to the complexity of multimodal integration and the scarcity of emotionally annotated data, reinforcing the relevance and potential contribution of the work.

The following Gantt diagram shows the established work plan.

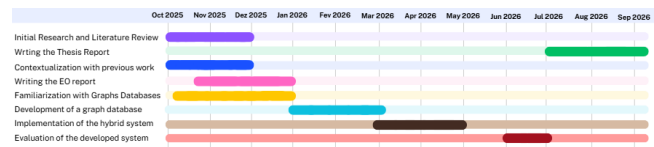


Figure 1. Project Timeline - Gantt Chart

no background falta falar da avaliações dos RS. Aqui debes mesmo dizer que métodos de avaliação se vao usar, tipo de split, etc

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